

AetherForge Compute Ltd

Powering Private Compute from the Land

Agrivoltaic solar + modular high-performance
computing on dual-use Devon farmland.
Behind-the-meter power. Resilient infrastructure.
Privacy-first workloads.

PILOT PHASE · DEVON, UK



A convergence of powerful structural tailwinds

Explosive AI & Compute Demand

AI training, inference, and privacy-preserving workloads are driving unprecedented demand for distributed, energy-efficient, high-performance compute. Traditional data centres face acute power, grid connection, and planning constraints.

UK Energy & Grid Pressure

Elevated industrial electricity prices and extended grid connection timelines make behind-the-meter generation and co-location exceptionally attractive. This model captures value that would otherwise be lost to curtailment or unfavourable export economics.

Land Use & Agricultural Innovation

Farmers and landowners face sustained margin pressure. Dual-use agrivoltaic schemes deliver new diversified revenue streams while preserving productive agricultural activity, biodiversity, and eligibility for support schemes.

Demand for Sovereign & Resilient Infrastructure

Growing requirement for private, censorship-resistant, and geographically sovereign compute capacity. Organisations increasingly seek alternatives to hyperscale cloud concentration with greater control over data, energy provenance, and operational resilience.

Agrivoltaic Solar-Powered Modular Compute Infrastructure

We develop pilot-scale facilities that combine solar energy generation with high-performance modular compute on the same land — maximising value per acre while preserving full agricultural use and delivering institutional-grade infrastructure.

01 Behind-the-Meter Solar

Ground-mounted arrays sized to match on-site compute demand. Maximises self-consumption and value capture while minimising grid export and connection dependencies.

02 Modular Compute Nodes

Purpose-built or containerised modules housing efficient, high-density hardware. Scalable, relocatable, with excellent thermal characteristics and physical security for rural deployment.

03 Dual-Use Agrivoltaic Land

Sheep grazing integrated beneath and between arrays. Maintains agricultural activity, supports planning arguments, enhances biodiversity and soil health.

04 Private & High-Value Workloads

Purpose-optimised for privacy-preserving compute, confidential AI inference, and other decentralised high-performance tasks prioritising sovereignty, efficiency and resilience.



A new revenue model that complements farming

Stable, Long-Term Income

Long-lease arrangements provide predictable, inflation-resilient revenue, diversifying farm income and reducing exposure to commodity price volatility and subsidy changes.

Maintain Agricultural Activity

Sheep grazing continues under the arrays. The land remains in productive agricultural use — a critical advantage in planning discussions and support scheme eligibility.

Low Visual & Operational Impact

Well-designed layouts with appropriate setbacks, screening, and professional management minimise landscape impact compared to pure solar or industrial developments.

Future-Proofing the Estate

Position land at the intersection of renewable energy and digital infrastructure — two of the strongest structural trends of the coming decades.

Planning & Community Benefits

Dual-use agrivoltaic schemes are increasingly viewed favourably by planning authorities when they deliver clear environmental, economic and community benefits.

Minimal Day-to-Day Burden

Professional management of compute and energy assets. Landowner involvement focused on lease terms and high-level oversight rather than operations.

An ideal environment for this model

Strong Solar Resource

Devon offers excellent solar irradiance for the UK. South-facing slopes and open land are well suited to ground-mount arrays with attractive capacity factors.

Available Rural Land

Mix of pasture and marginal arable land suitable for dual-use. 10–25 acre parcels are realistic targets near Okehampton, Ivybridge, and surrounding areas.

Planning Context

Devon authorities are experienced with renewable energy applications. Agrivoltaic and farm diversification schemes have established precedent and policy support.

Connectivity

Starlink provides excellent low-latency backhaul for rural sites today. Future fibre and 5G improvements will further strengthen the proposition.

Skilled Local Base

Proximity to Plymouth and Exeter for technical support, contractors, and talent. Strong engineering and agricultural heritage in the region.

Aligned Values

Devon's culture of practical self-reliance, land stewardship, and measured innovation makes it a natural home for resilient, decentralised infrastructure.

Conservative base case from detailed financial model (May 2026)

Interest-only debt for first 5 years | 320 kWp solar system

CAPITAL STRUCTURE

Angel Equity (founder + investor)	£250,000
Green Loan (7.5% interest-only)	£225,000
Land Debt (10% interest-only)	£81,000
Hardware Debt (10% interest-only)	£280,500
Total Debt (interest-only, 5 yrs)	~£586,500

SYSTEM SCALE & OPERATIONS

Core Compute Load	11,500 kWh/month
AI/HPC Premium Load	12,000 kWh/month
Total On-Site Draw	23,500 kWh/month
Operating Costs (full run rate)	£2,000/month
Monthly Debt Service	~£4,419
Stabilised Monthly Revenue	£20k – £21.5k

SCENARIO ANALYSIS (5-Year Equity Returns)

Scenario	Equity Multiple	Cash Recovery	Min Cumulative Cash	Notes
Base Case	2.71x	Jul 2028	£26.2k	Current model
Lower AI demand (-25%)	1.90x	Mar 2029	£22.3k	Lower AI/HPC revenue
Combined Downside	1.69x	Jul 2029	£17.8k	-20% AI, -30% base, +20% opex
Higher AI demand (+25%)	3.52x	Mar 2028	£30.0k	Stronger AI/HPC uptake

Key: All scenarios remain cash positive after initial ramp-up. Returns highly sensitive to AI/HPC revenue share and site costs. Pilot designed for replicability.

Proactive risk management for a resilient pilot

RISK

Planning Permission

MITIGATION

Early engagement with Devon planning officers. Strong agrivoltaic precedent and dual-use agricultural narrative. Professional planning support from day one.

RISK

Revenue & Energy Price Volatility

MITIGATION

Conservative modelling with diversified workload mix. Focus on efficient hardware. Potential for long-term offtake agreements. Structural value in energy transition remains robust.

RISK

Hardware & Technology Risk

MITIGATION

Modular, proven containerised designs. Staged deployment. Focus on mature, efficient hardware platforms with strong secondary markets and upgrade paths.

RISK

Grid Connection & Infrastructure

MITIGATION

Behind-the-meter design minimises export reliance. Early DNO engagement. Starlink as primary resilient backhaul with fibre fallback.

RISK

Regulatory & Perception Risk

MITIGATION

Strong agrivoltaic + continued sheep grazing framing. Transparent community engagement. Emphasis on dual-use agricultural and local economic benefit.

RISK

Operational & Security

MITIGATION

Professional remote monitoring and 24/7 oversight. Physical security design. Experienced partners for operations. Comprehensive insurance and maintenance contracts.

What we are looking for

The pilot offers a de-risked, capital-efficient entry point into a scalable platform at the intersection of AI infrastructure demand and the energy transition. We are seeking aligned partners to accelerate deployment in Devon.

Land Partners

Landowners or agents with suitable 10–25 acre sites in Devon (particularly around Okehampton, Ivybridge, South Hams). Long-term lease interest with dual-use agrivoltaic model.

Planning & Development Support

Introduction to planning consultants and early conversations with local authorities. Help shaping applications that emphasise agricultural continuity, biodiversity net gain, and community benefit.

Early Technical Partners

Engineering, electrical, and solar installation partners interested in innovative rural infrastructure projects. Container and hardware suppliers with proven rural deployment experience.

Strategic & Capital Partners

Investors or organisations aligned with decentralised infrastructure, climate tech, and the energy transition who see the long-term platform potential of integrated agrivoltaic compute at scale.

AetherForge Compute Ltd

Powering Private Compute from the Land

Pilot Phase — Devon, United Kingdom

We are building the next generation of resilient, land-integrated compute infrastructure.

This pilot represents a compelling, de-risked opportunity to participate in the convergence of AI demand and decentralised energy systems.

For further discussion or to explore partnership opportunities:

Joshua Karkeek | Director
AetherForge Compute Ltd
joshua@aetherforgecompute.com

CONFIDENTIAL